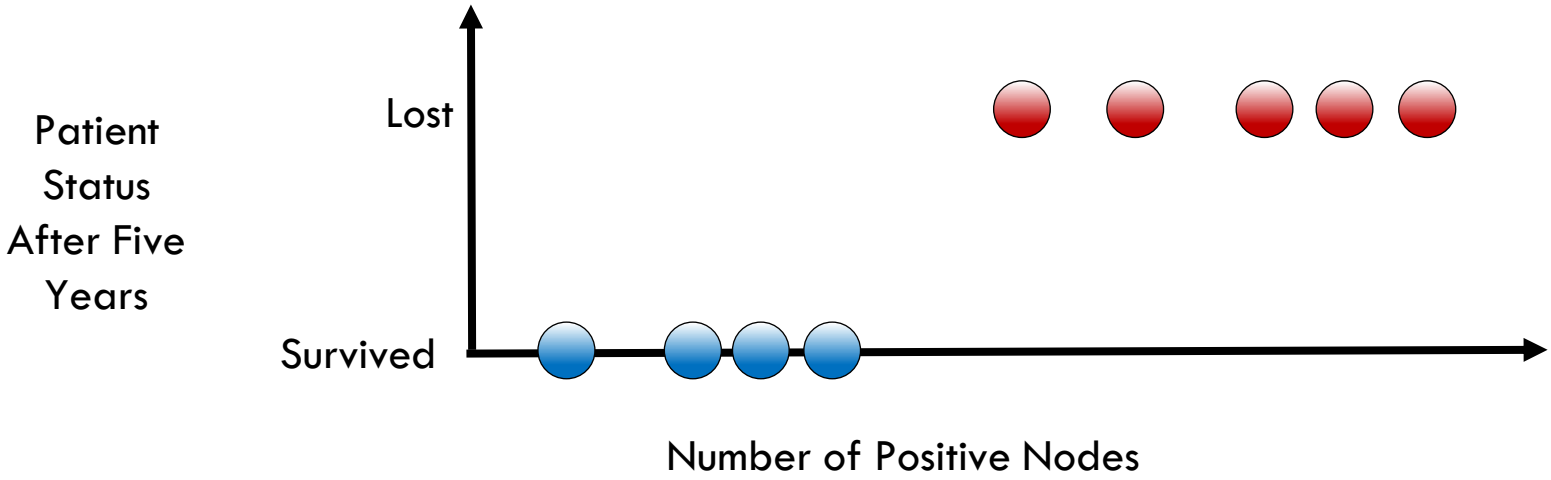
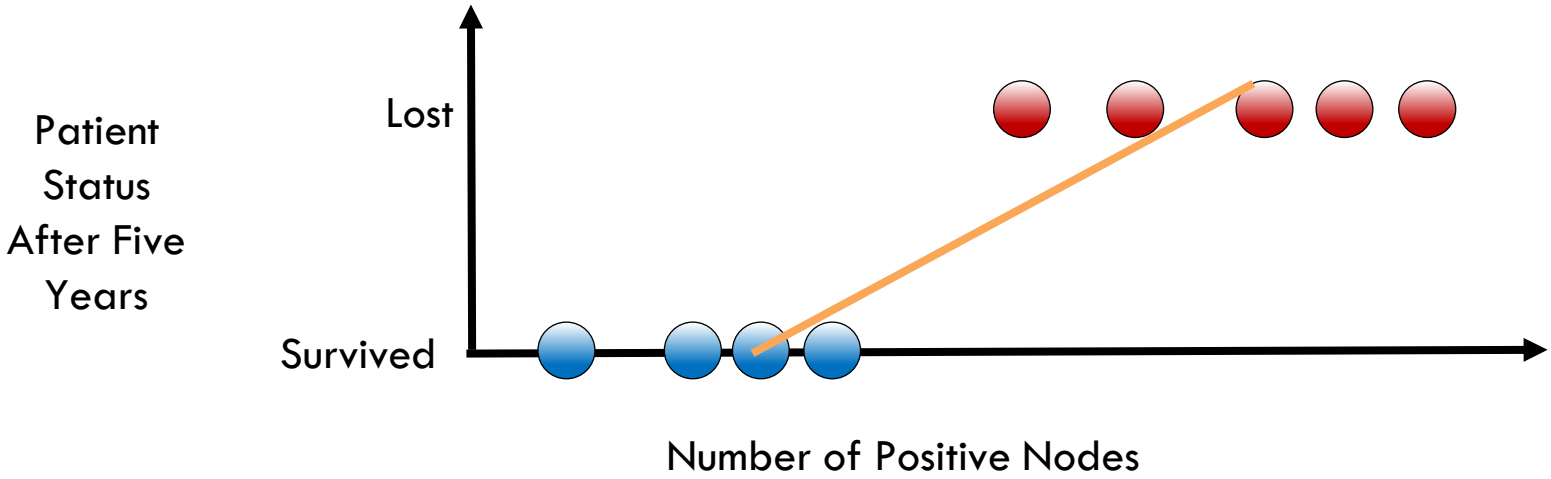
Logistic Regression

Introduction to Logistic Regression



Linear Regression for Classification?



$$y_\beta(x) = \beta_0 + \beta_1 x + \varepsilon$$

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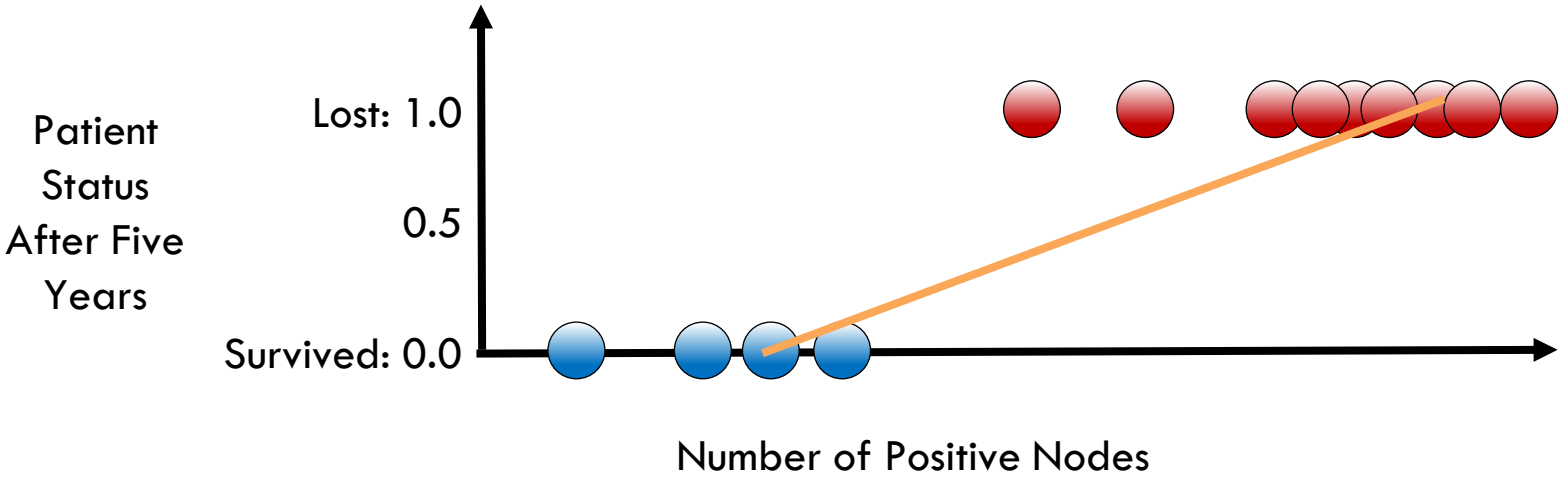

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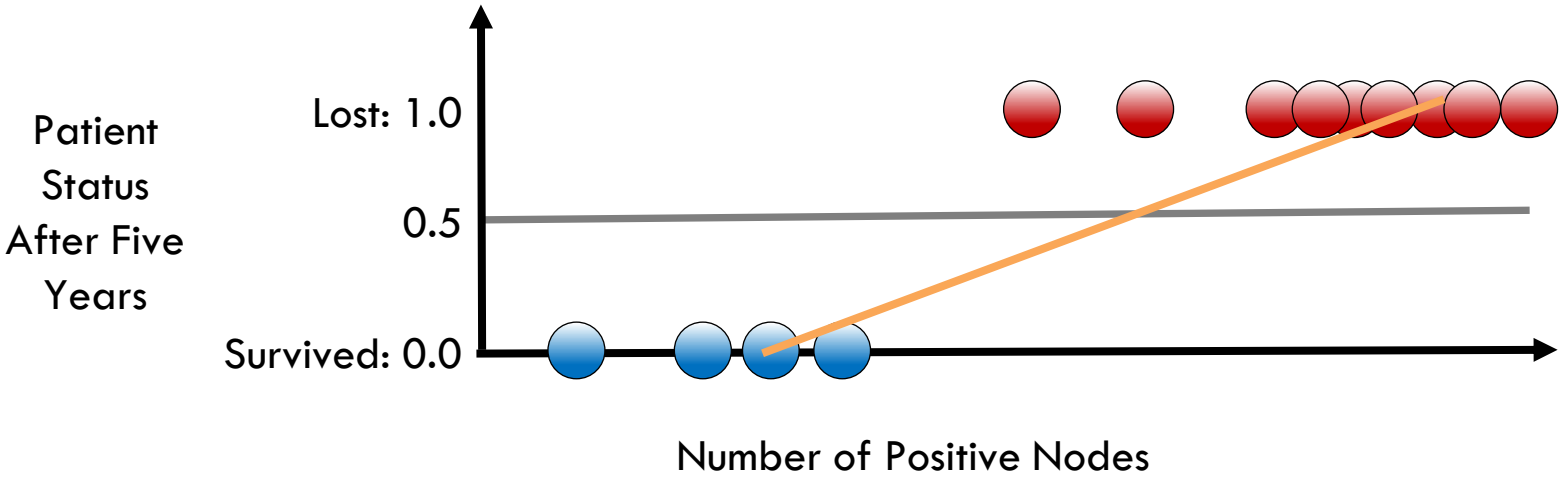

If model result > 0.5: predict lost
If model result < 0.5: predict survived

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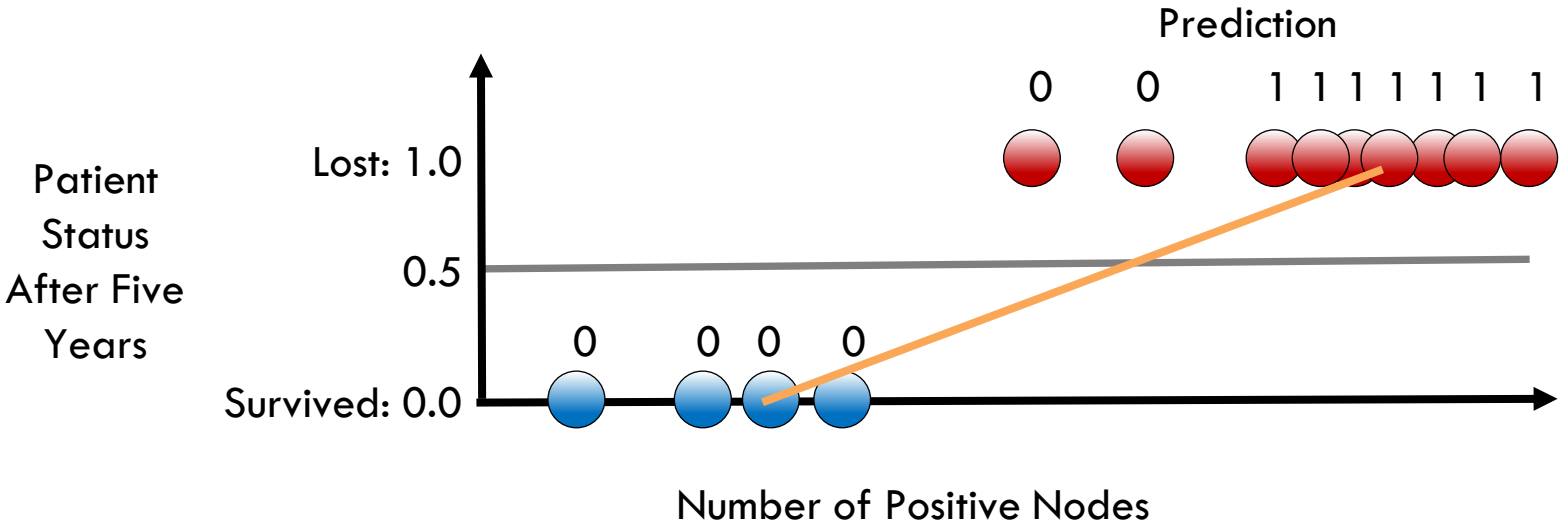

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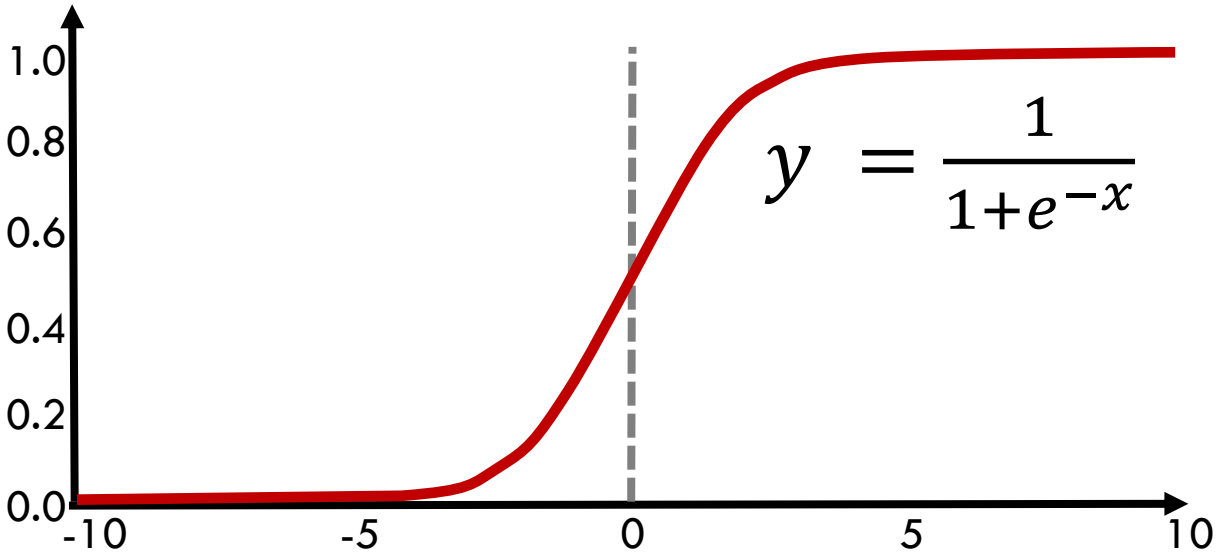
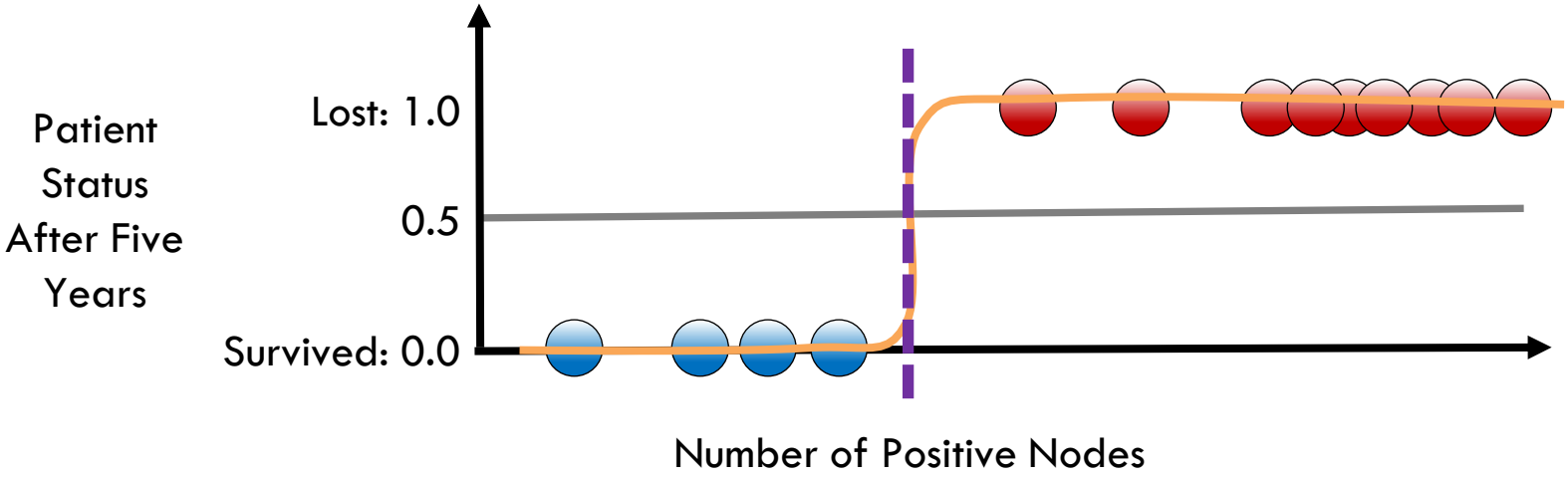
What is this Function?

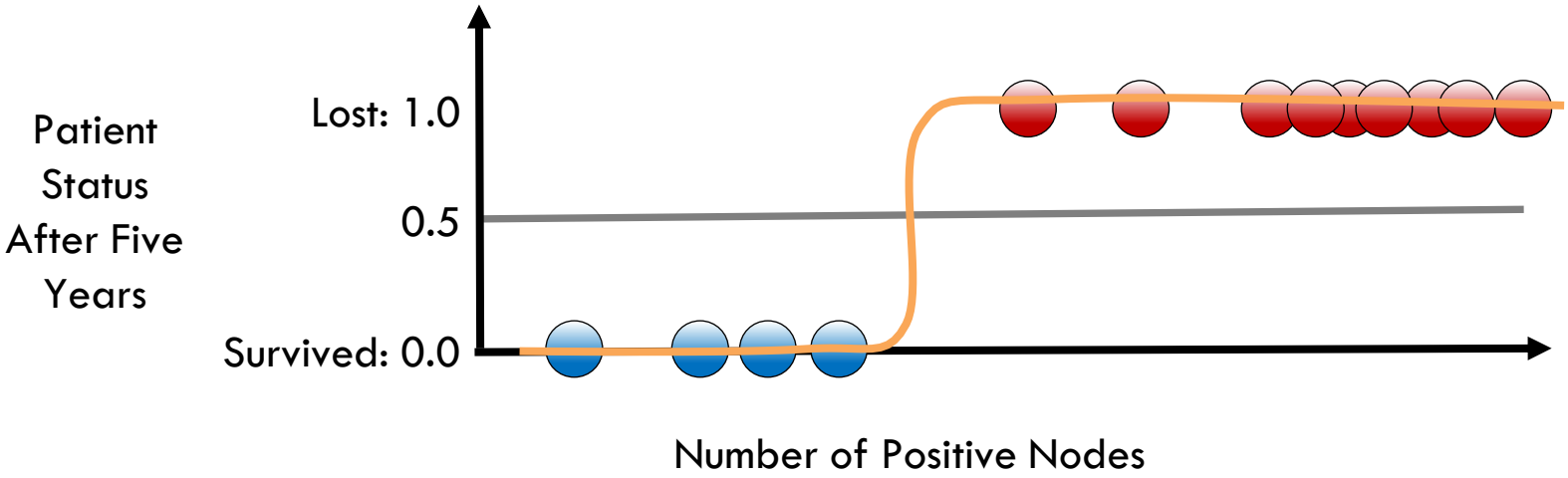


The Decision Boundary



$$y_\beta(x) = \frac{1}{1+e^{-(\beta_0+\beta_1 x+\varepsilon)}}$$

Logistic Regression



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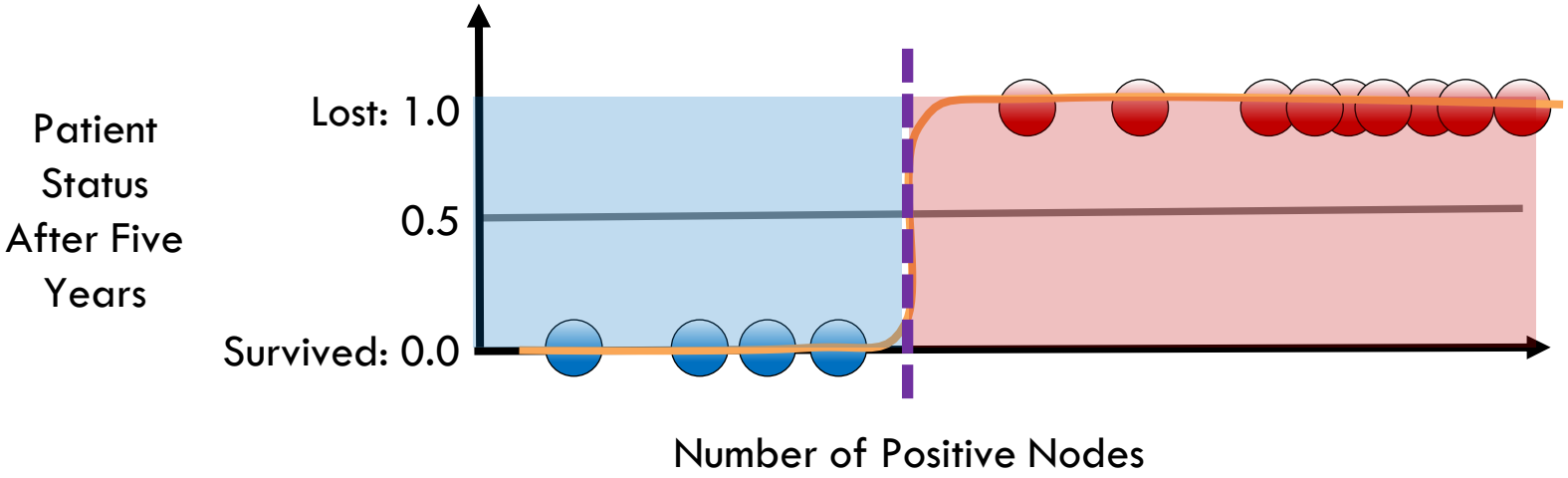

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Relationship of Logistic to Linear Regression

Logistic Function

$$P(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x + \varepsilon)}}$$

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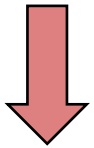

Odds Ratio

$$\frac{P(x)}{1 - P(x)} = e^{(\beta_0 + \beta_1 x)}$$

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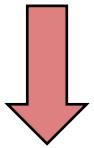

Log Odds

$$log\left[\frac{P(x)}{1 - P(x)}\right] = \beta_0 + \beta_1 x$$

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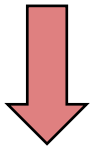

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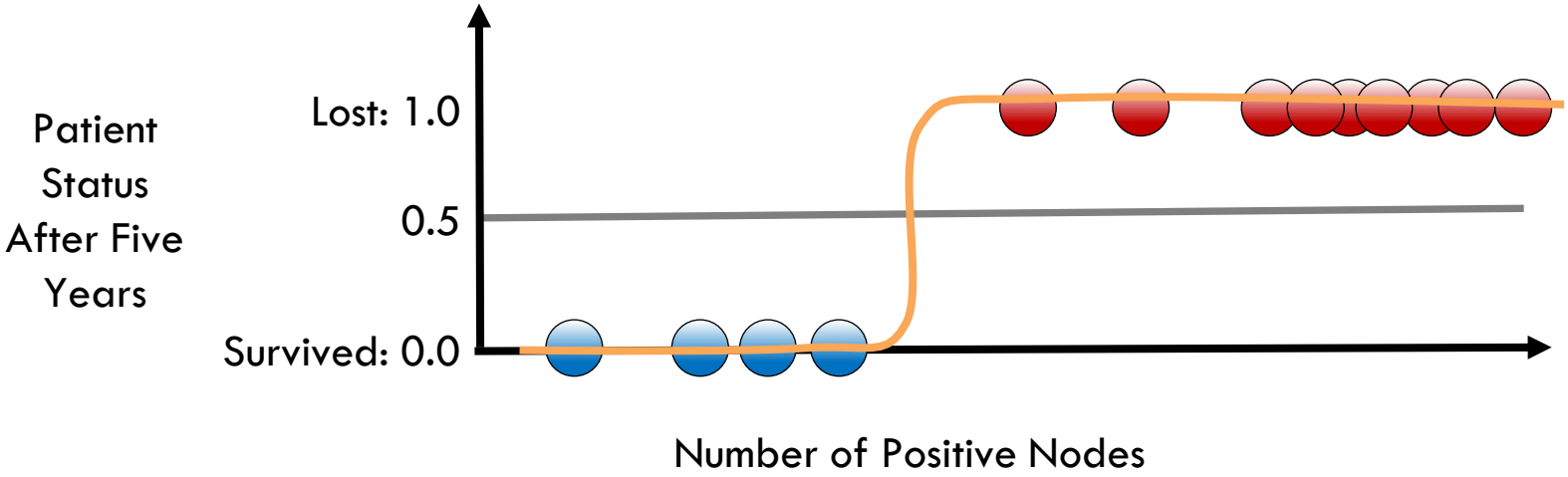
Classification with Logistic Regression

One feature (nodes)
Two labels (survived, lost)



Classification with Logistic Regression

Two features (nodes, age)
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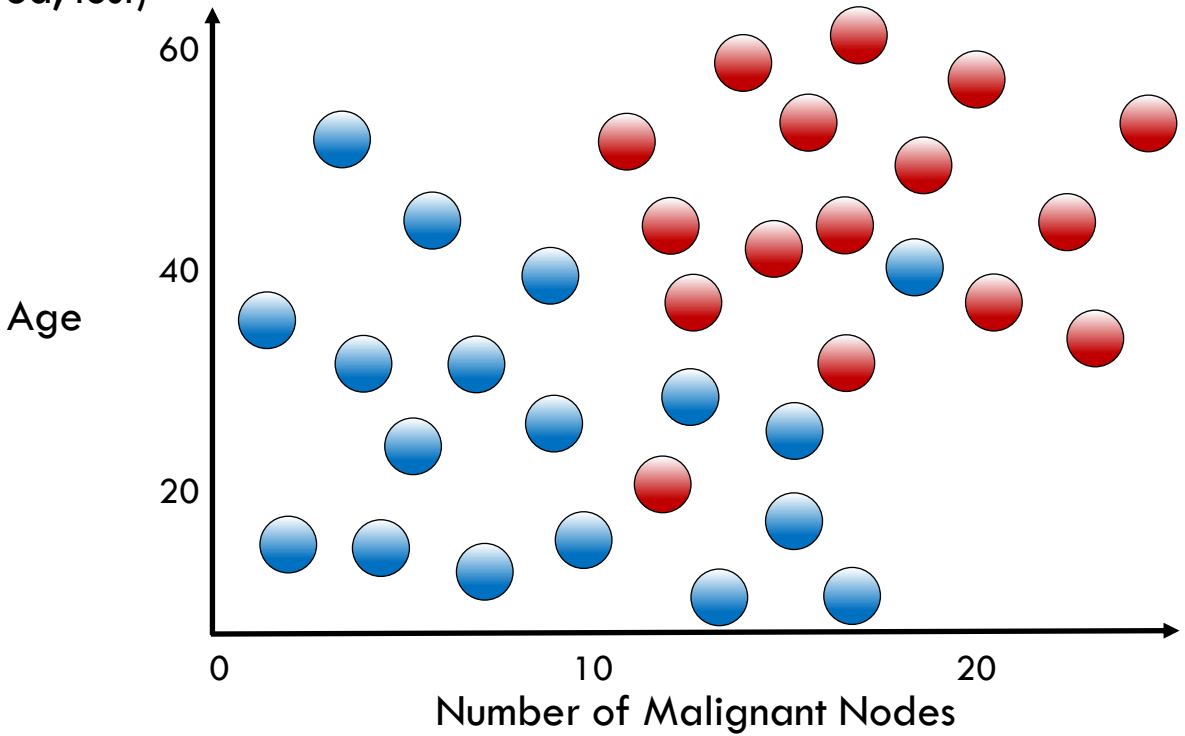

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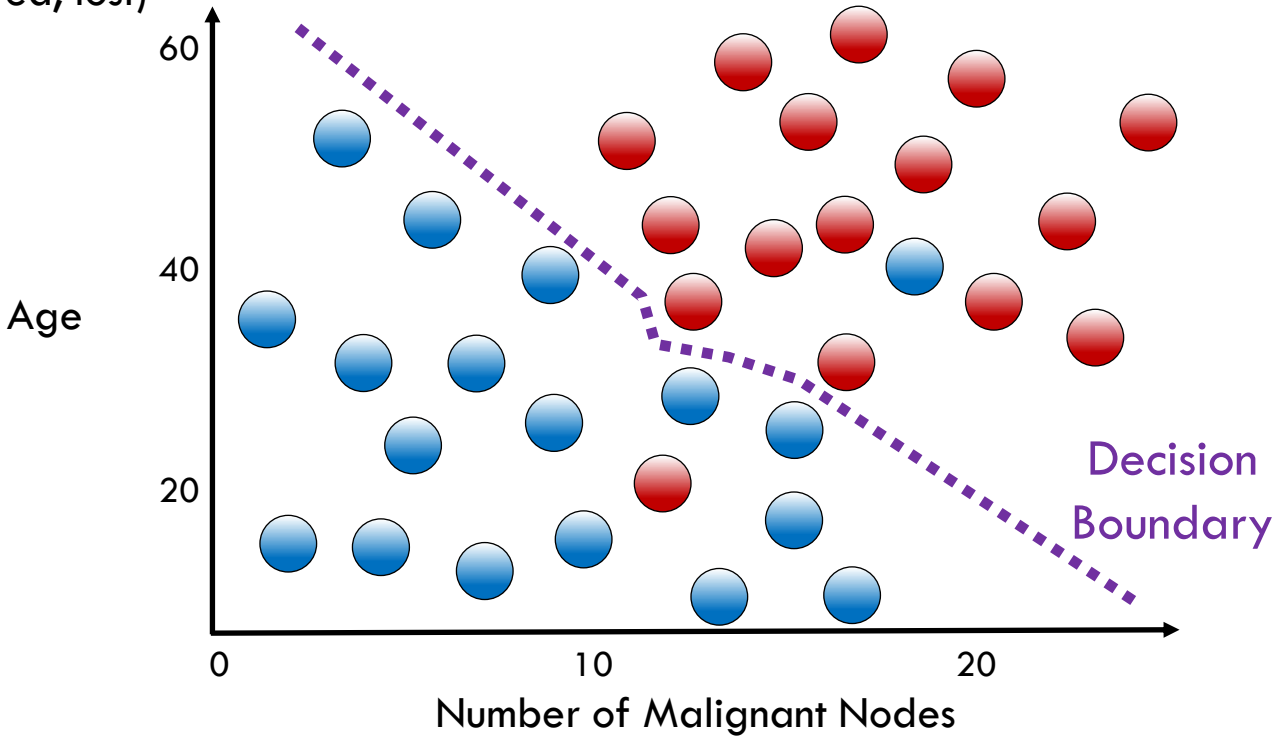

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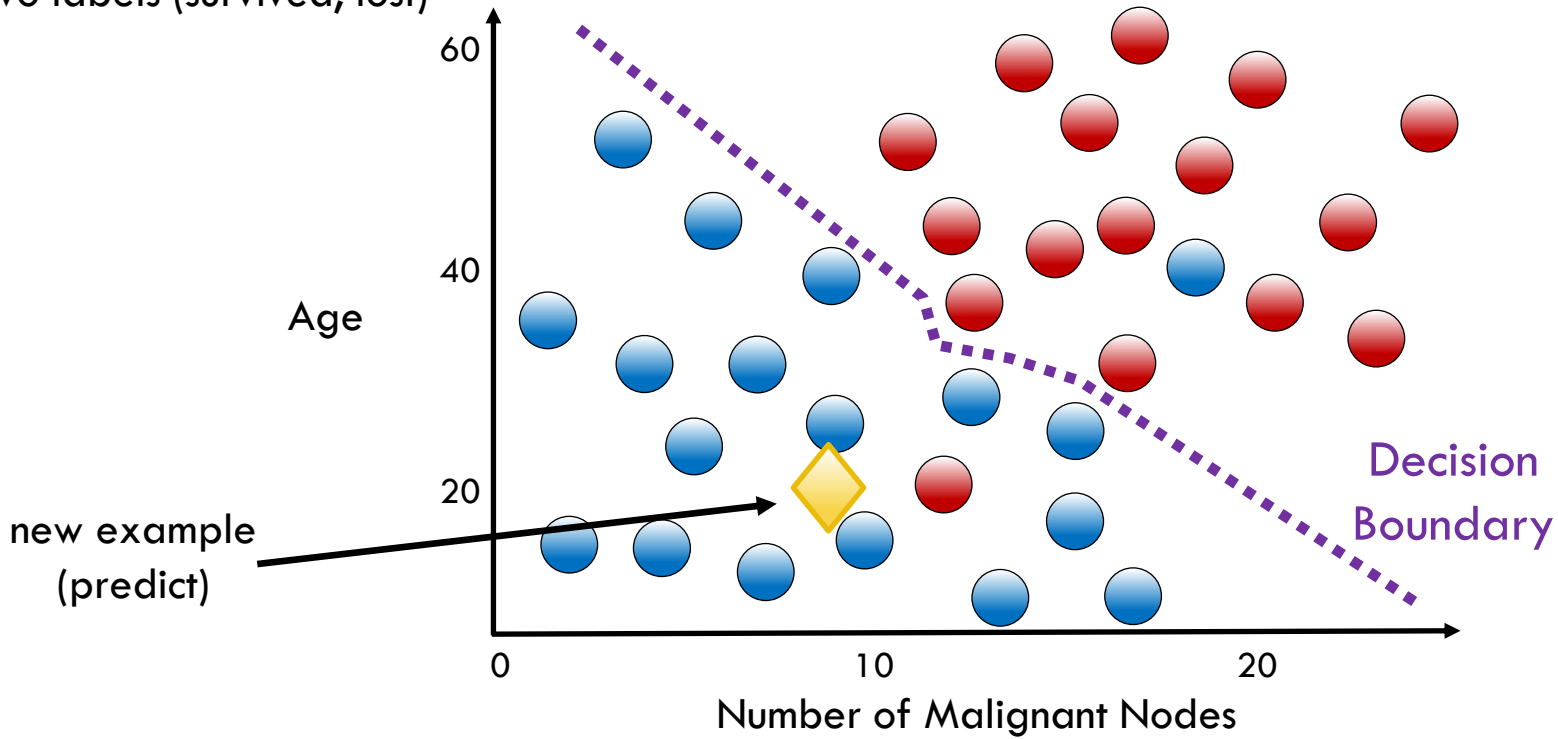
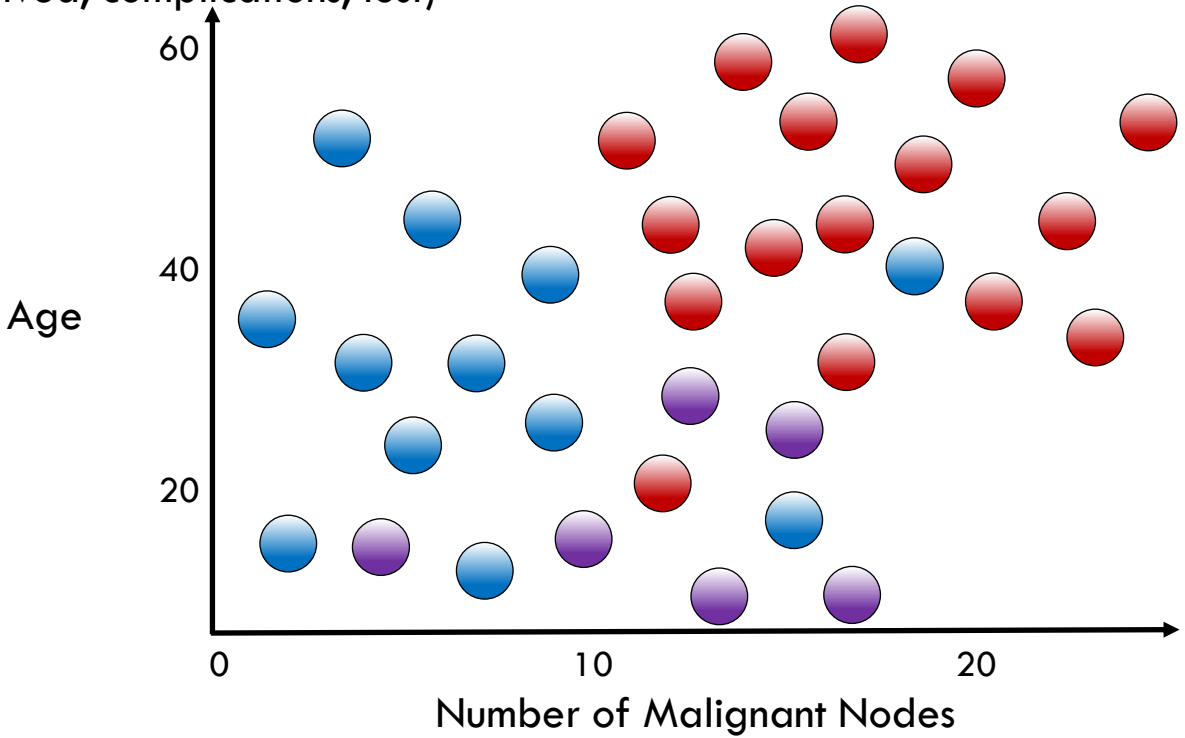


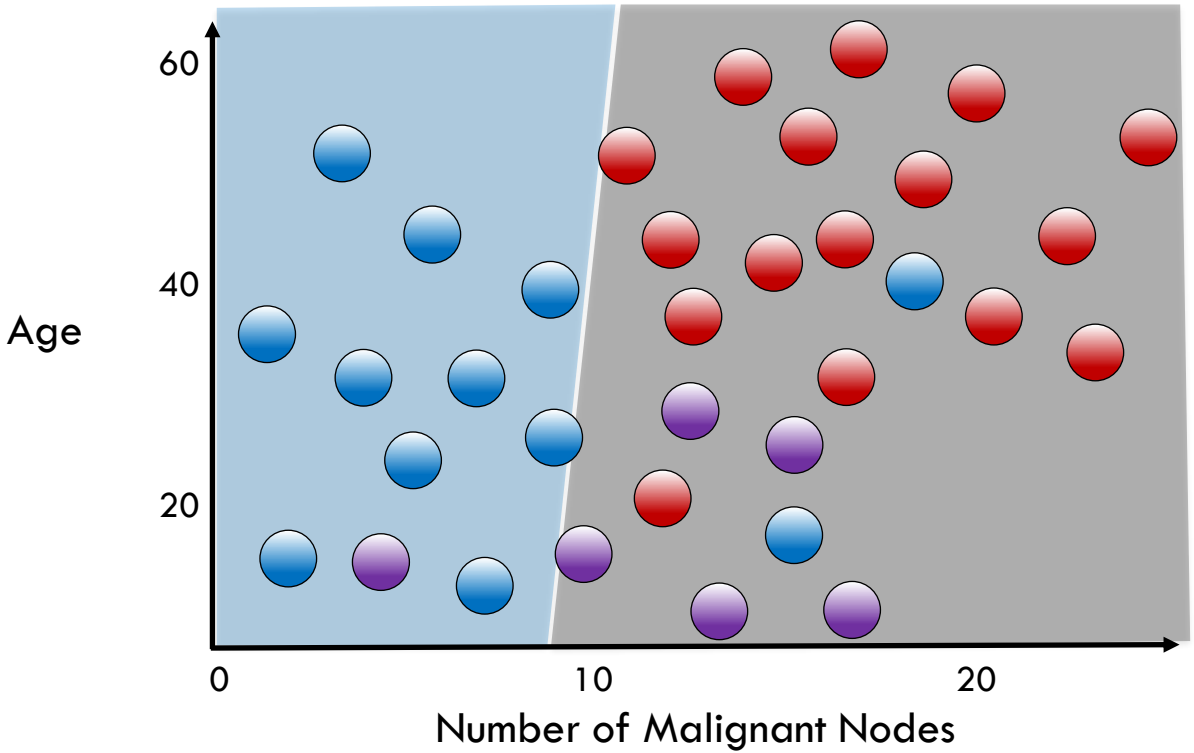
Multiclass Classification with Logistic Regression

Two features (nodes, age)

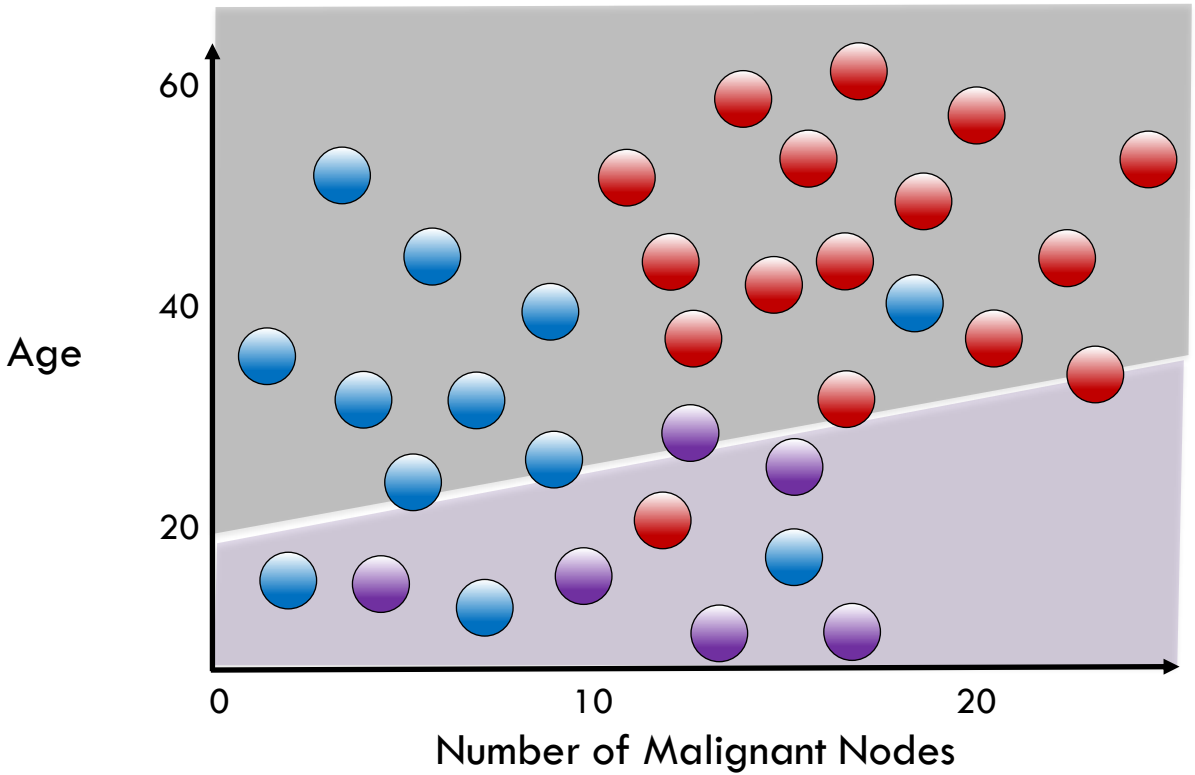
Three labels (survived, complications, lost)



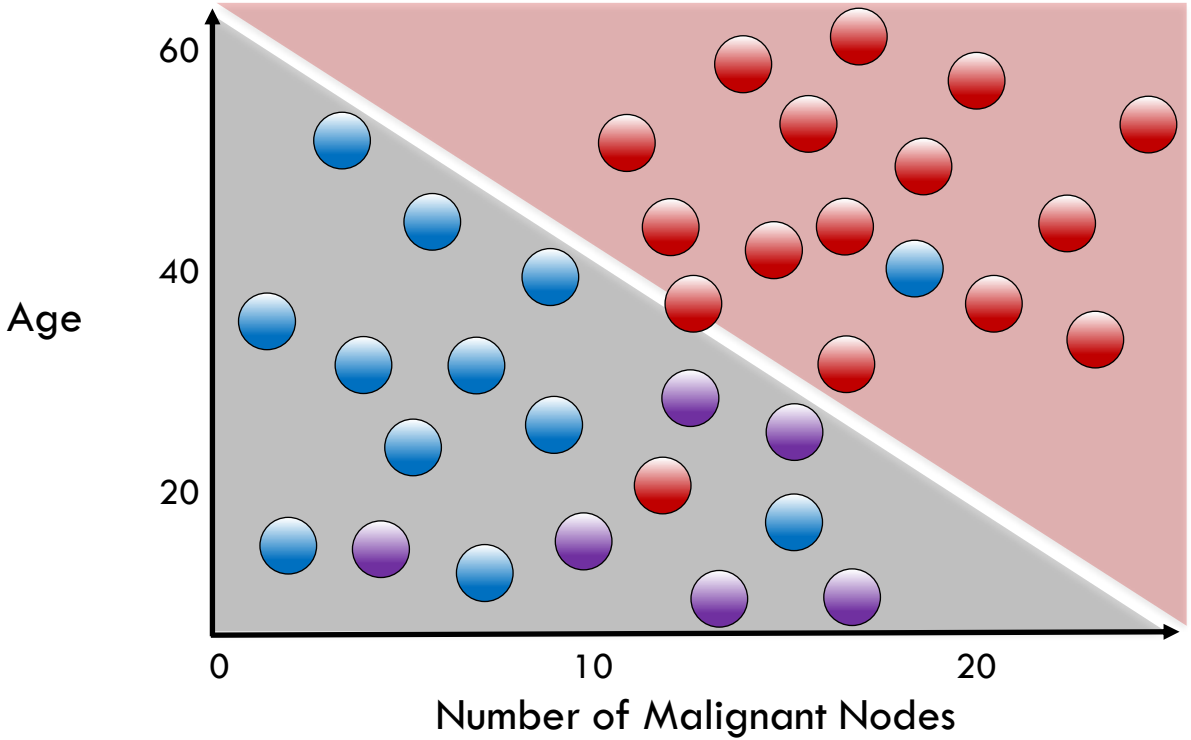
One vs All: Survived vs All



One vs All: Complications vs All

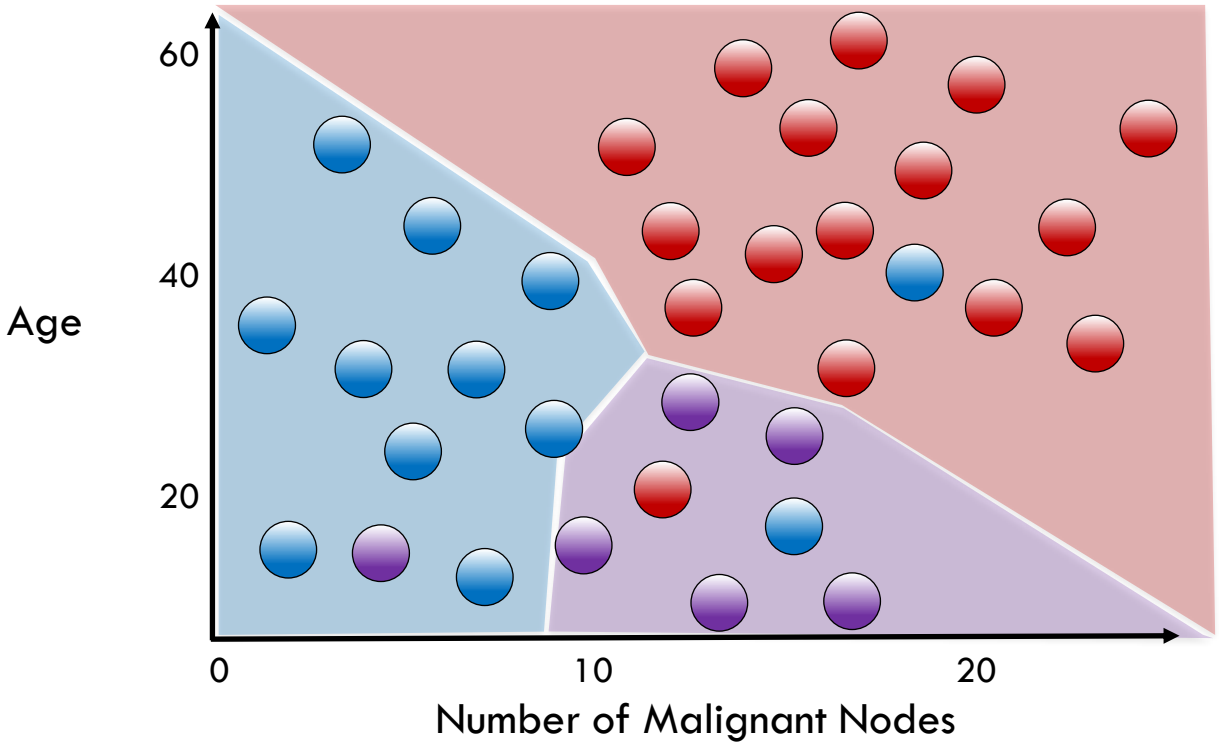


One vs All: Loss vs All



Multiclass Decision Boundary

Assign most probable class to each region



Logistic Regression: The Syntax

Import the class containing the classification method

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from sklearn.linear_model import LogisticRegression
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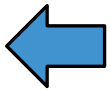

regularization parameters

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Tune regularization parameters with cross-validation: LogisticRegressionCV.